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Option hedging theory under transaction costs

Tze Leung Lai [a,1], Tiong Wee Lim [b,*,2]

[a] Department of Statistics, Sequoia Hall, Stanford University, Stanford, CA 94305-4065, USA
[b] Department of Statistics and Applied Probability, National University of Singapore, 6 Science Drive 2, Singapore 117546, Republic of Singapore

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ABSTRACT

The problem of option hedging in the presence of proportional transaction costs can be formulated as a singular stochastic control problem. Hodges and Neuberger [1989. Optimal replication of contingent claims under transactions costs. Review of Futures Markets 8, 222–239] introduced an approach that is based on maximization of the expected utility of terminal wealth. We develop a new algorithm to solve the corresponding singular stochastic control problem and introduce a new approach to option hedging which is closer in spirit to the pathwise replication of Black and Scholes [1973. The pricing of options and corporate liabilities. Journal of Political Economy 81, 637–654]. This new approach is based on minimization of a Black–Scholes-type measure of pathwise risk, defined in terms of a market delta, subject to an upper bound on the hedging cost. We provide an efficient backward induction algorithm for the problem of cost-constrained risk minimization, whose associated singular stochastic control problem is shown to be equivalent to an optimal stopping problem. This algorithm is then modified to solve the singular stochastic control problem associated with utility maximization, which cannot be reduced to an optimal stopping problem. We propose to choose an optimal parameter (risk-aversion coefficient or Lagrange multiplier) in either approach by minimizing the mean squared hedging error and demonstrate that with this "best" choice of the parameter, both approaches have similar performance. We also discuss the different notions of risk in both approaches and propose a volatility adjustment for the risk-minimization approach, which is analogous to that introduced by Zakamouline [2006. European option pricing and hedging with both fixed and proportional transaction costs. Journal of Economic Dynamics and Control 30, 1–25] for the utility maximization approach, thereby providing a unified treatment of both approaches.

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1. Introduction

The problem of option pricing and hedging was initially studied in an idealized setting where an investor incurs no transaction costs from trading in a market consisting of a risk-free asset ("bond") with constant rate of return and a risky asset ("stock") whose price is a geometric Brownian motion with constant rate of return and volatility. For this setting, Black and Scholes (1973) demonstrated that in the absence of arbitrage the value of an option is an expectation of the

* Corresponding author. Tel.: +65 6516 7857; fax: +65 6872 3919.
E-mail addresses: lait@stanford.edu (T.L. Lai), stalimtw@nus.edu.sg (T.W. Lim).
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discounted payoff at expiration under the "risk-neutral" measure, for which the stock's rate of return equals the risk-free rate. Moreover, perfect replication of the option is possible and the option is itself "redundant" in such a "complete" market. However, the Black–Scholes "delta-hedging" portfolio requires continuous trading. In the presence of transaction costs proportional to the amount of trading, such a continuous strategy is prohibitively expensive. Hence it is impossible to perfectly replicate the option in this setting when there are transaction costs and, as a result, trading in an option involves an essential element of risk.

One approach to characterize this hedging risk examines the difference between the realized cash flow from a hedging strategy and the desired payoff at maturity. It embeds option hedging within the framework of portfolio selection introduced by Magill and Constantinides (1976) and Davis and Norman (1990), and uses a risk-averse utility function to assess this shortfall ("replication error"). In this way, Hodges and Neuberger (1989) formulated the problem of option hedging as that of maximizing the investor's expected utility of terminal wealth. Making use of an indifference argument, the reservation selling or buying price of an option is defined as the amount of money that would make an investor indifferent, in terms of expected utility, between trading in the market with and without a (short or long) position in the option. This involves the value functions of two singular stochastic control problems and the optimal hedge is given by the difference in the trading strategies corresponding to these two problems. The nature of the optimal hedge is that an investor with an option position should rebalance his portfolio only when the number of shares of stock falls "too far" out of line. For the negative exponential utility function, Davis et al. (1993), Clewlow and Hodges (1997) and Zakamouline (2006) have developed numerical methods to compute the optimal hedge and option price by making use of discrete-time dynamic programming for an approximating binomial tree for the stock price. Whalley and Wilmott (1997) and Barles and Soner (1998) have developed asymptotic approximations for these hedging strategies and option prices as the transaction costs approach 0. Constantinides and Zariphopoulou (1999, 2001) have provided option price bounds under general utility functions (rather than the negative exponential utility function commonly adopted for numerical studies). In this paper we make use of a new numerical method for solving singular stochastic control problems, recently introduced by Lai et al. (2009), to develop a much simpler algorithm to compute the buy–sell boundaries and value functions in the utility-based approach.

In the presence of transaction costs, alternatives to the utility-based approach have been based on super-replication (or replication) in a discrete-time setting and are concerned with finding trading strategies which produce payoffs at expiration that are at least (or exactly) as valuable as the option payoff. Noting that using the Black–Scholes delta to short-sell delta shares of stock at the beginning of each revision interval introduces too high transaction costs as the width of the revision interval shrinks to 0, Leland (1985) proposed a modification of the variance used in the Black–Scholes delta so as to yield the desired option payoff at expiration inclusive of transaction costs. The fact that this modified strategy is not self-financing has prompted Boyle and Vorst (1992) to work in a discrete-state (binomial tree) framework to construct a *self-financing* discrete-time replicating strategy, thereby extending the two-period model of Merton (1990, Chapter 14). Explicit portfolio weights at each node of the binomial tree can be computed by using a backward induction procedure. However, these methods require the user to exogenously specify a revision interval and it is unclear how one can do so optimally. In fact, as the width of the revision interval approaches 0, the cost of the option approaches the price of a single share of stock, which turns out to be the least expensive way of super-replicating the option in a continuous-time model; see Soner et al. (1995). For the binomial tree model, Bensaid et al. (1992) derived bounds on the option value at inception by minimizing the initial cost of the self-financing strategy used to produce a super-replicating portfolio of stock and bond at expiration. As they have shown, by rebalancing only in the earlier periods, it is possible to have a super-replicating portfolio that is *less expensive* than the corresponding replicating portfolio. In general, the optimal discrete-time super-replicating strategy is such that the investor with an option position does not transact at a trading date if the inherited amount of stock is in a certain range (which depends on the past history of the stock price); otherwise he adjusts his portfolio back to this range. Noting that this cost minimization problem associated with super-replication is path dependent and that the dynamic programming algorithm is computationally expensive if the number of periods is not sufficiently small, Edirisinghe et al. (1993) developed a linear programming algorithm and a two-stage dynamic programming method to approximate the optimal solution. More recently, Primbs (2009) provided an alternative formulation of super-replication in terms of the first two moments of the replication error.

In this paper we propose a new approach which formulates the option hedging problem in the presence of transaction costs as a constrained risk minimization problem that minimizes a measure of pathwise risk under the constraint that the hedging cost (central to the replication/super-replication approach) does not exceed a prescribed level. This measure of pathwise risk is implicit in the Black–Scholes theory that continuously rebalances the portfolio to make such risk zero when there are no transaction costs, and leads in our approach to a natural modification of the Black–Scholes delta-hedging scheme. In the presence of transaction costs, this modification consists of buying or selling the underlying stock whenever the holding of shares falls outside a no-transaction band containing the option's delta. The corresponding singular stochastic control problem, whose no-action region is the no-transaction band, is equivalent to an optimal stopping problem. This equivalence is used to compute the buy- and sell-boundaries efficiently, thereby reducing substantially the computational complexity of the original singular stochastic control problem, which requires determination of both *when* to apply the control (in the form of buying or selling) and *how much* control to apply.

This paper is organized as follows. Section 2 defines the hedging cost of a self-financing strategy and uses it to formulate the singular stochastic control problems associated with the utility maximization and the cost-constrained risk

minimization approaches to option hedging in the presence of transaction costs. It will be shown that both formulations of the option hedging problem belong to the general framework of minimizing expected hedging cost under a risk constraint. Section 3 presents new algorithms, which are considerably simpler than those currently available, to compute the optimal buy- and sell-boundaries for both singular stochastic control problems. Section 4 gives some numerical results to illustrate the computational schemes and compare their hedging errors with those of Leland (1985) and Black and Scholes (1973) with prescribed revision intervals. Some concluding remarks are given in Section 5.

2. Singular stochastic control in option hedging with transaction costs

Suppose an investor is presented with an opportunity to enter into a position in a European call option written on a stock, with strike price K and expiration date T. The stock price process is assumed to follow a geometric Brownian motion with mean rate of return $\alpha > 0$ and volatility $\sigma > 0$:

$$dS_t = \alpha S_t\, dt + \sigma S_t\, dW_t, \tag{1}$$

where $\{W_t,\ t \geq 0\}$ is a standard Brownian motion with $W_0 = 0$. In the presence of proportional transaction costs, the investor pays fractions $0 < \lambda < 1$ and $0 < \mu < 1$ of the dollar value transacted on purchase and sale of the underlying stock, respectively. Let x_t denote the number of shares held in stock and y_t the dollar value of investment in bond, which pays a fixed risk-free rate $r > 0$. The investor's position (x_t, y_t) in stock and bond is given by

$$dx_t = dL_t - dM_t \quad (\text{equivalently, } x_t = L_t - M_t), \tag{2a}$$

$$dy_t = ry_t\, dt - aS_t\, dL_t + bS_t\, dM_t \tag{2b}$$

with $a = 1 + \lambda$ and $b = 1 - \mu$, where (L, M) is a pair of non-decreasing and non-anticipating processes such that L_t (resp. M_t) denotes the cumulative number of shares of stock bought (resp. sold) within the time interval $[0, t]$, $0 \leq t \leq T$. With $i = 0, s$ and b indicating the investor's position in the option (no call option, short call and long call, respectively), the terminal wealth of the investor is $\Omega_T^i = y_T + Z^i(S_T, x_T)$, where $Z^i(S, x)$ represents the terminal settlement value of the stock and option (the latter applies for $i = s, b$ only). When there is no option position ($i = 0$), the liquidated value of stock is given by

$$Z^0(S, x) = xS(a\mathbb{1}_{\{x<0\}} + b\mathbb{1}_{\{x\geq 0\}}). \tag{3}$$

If the option is *asset settled*, then the writer delivers one share of stock in return for a payment of K when the buyer exercises the option at maturity T, so

$$Z^i(S, x) = Z^0(S, x - \Delta^i(S)) + K\Delta^i(S), \quad i = s, b, \tag{4}$$

where $\Delta^s(S) = \mathbb{1}_{\{S>K\}}$ (short call) and $\Delta^b(S) = -\mathbb{1}_{\{S>K\}}$ (long call). In the case of *cash settled* options, the writer delivers $(S_T - K)^+$ in cash at T, so

$$Z^i(S, x) = Z^0(S, x) - (S - K)\Delta^i(S), \quad i = s, b. \tag{5}$$

Implicit in definitions (3)–(5) are the following terminal trades in stock: for $i = 0$ and s, b with cash settlement, trade to 0 share of stock; while for $i = s, b$ with asset settlement, trade to $\Delta^i(S_T)$ share of stock. Although we only consider call options on stocks that pay no dividends in the sequel, a straightforward modification of the arguments yields corresponding results for European put and call options on dividend-paying stocks, as will be summarized in Section 5.

2.1. Hedging cost of a self-financing trading strategy

A quantity that is of basic importance in option hedging theory is the *hedging cost* incurred over any given finite horizon $[t, T]$, given by

$$C^i(L, M; t, T) = \Psi(L, M; t, T) - Z^i(S_T, x_T), \quad i = 0, s, b \tag{6}$$

as will be derived below, where

$$\Psi(L, M; t, T) = a\int_{[t,T)} e^{r(T-u)} S_u\, dL_u - b\int_{[t,T)} e^{r(T-u)} S_u\, dM_u \tag{7}$$

is the total trading cost (inclusive of transaction fees) incurred over $[t, T)$. Since the terminal wealth of the investor is

$$\Omega_T^i := y_T + Z^i(S_T, x_T) = y_t e^{r(T-t)} - C^i(L, M; t, T), \tag{8}$$

where y_t denotes the initial endowment at time t, the hedging cost is the difference between the terminal value of the initial endowment and the terminal wealth. For $i = s, b$, the terminal wealth can be interpreted as the *hedging error* (or replication error) of the strategy (x_t, y_t), $0 \leq t \leq T$, since it measures the *shortfall* between the liquidated value of the hedging (or replicating) portfolio and the desired option payoff.

One way to derive (6) is to extend the self-financing argument of Bensaid et al. (1992) to continuous time and continuous state. Specifically, the investor can choose to trade at any time in the interval $[t,T]$, and suppose that the first trade after time t occurs at time τ, so $x_u = x_t$ for all $u \in [t,\tau)$. The trade at τ incurs a cost of $S_\tau(a\,dL_\tau - b\,dM_\tau)$, since the investor can either purchase stock or sell stock, but not both. In the time that has elapsed, the risk-free account changes from y_t to $y_t e^{r(\tau-t)}$. Therefore, the new value y_τ in the risk-free account after the trade at time τ is

$$y_\tau = y_t e^{r(\tau-t)} - S_\tau(a\,dL_\tau - b\,dM_\tau).$$

More generally, suppose trades after time $\tau_0 = t$ occur at times $\tau_1 < \tau_2 < \cdots < \tau_n$, which are the jump times of (L_t, M_t) up to time T. It follows by induction that

$$y_T = y_t e^{r(T-t)} - \sum_{i=0}^{n} e^{r(T-\tau_i)} S_{\tau_i}(a\,dL_{\tau_i} - b\,dM_{\tau_i}) = y_t e^{r(T-t)} - \Psi(L,M;t,T),$$

which also follows directly from (2b) by integration. Since this value y_T does not include the proceeds (or losses) $Z^i(S_T, x_T)$ that arise from the terminal settlement of stock and option (if present), and since y_t does not depend on the particular trading strategy implemented after time t, the hedging cost incurred over $[t,T]$ is equal to (6).

As a special case, consider trading strategies which necessitate an instantaneous change in the amount of shares of stock whenever the stock position at time t moves outside some allowable range $[X_b(t,S_t), X_s(t,S_t)]$:

$$dL_t = (X_b(t,S_t) - x_{t-})^+ \quad \text{and} \quad dM_t = (X_s(t,S_t) - x_{t-})^+.$$

The corresponding stock position is given by

$$x_t = \begin{cases} X_b(t,S_t) & \text{if } x_{t-} < X_b(t,S_t), \\ X_s(t,S_t) & \text{if } x_{t-} > X_s(t,S_t), \\ x_{t-} & \text{if } X_b(t,S_t) \le x_{t-} \le X_s(t,S_t). \end{cases} \tag{9}$$

For such strategies, the total trading cost (7) can be expressed explicitly in terms of the trading boundaries X_b and X_s as

$$\Psi(L,M;t,T) = \int_{[t,T)} e^{r(T-u)} H(u, S_u, x_{u-})\,du, \tag{10}$$

where

$$H(t,S,x) = S[a(X_b(t,S) - x)^+ - b(x - X_s(t,S))^+]. \tag{11}$$

Note that $H(u, S_u, x_{u-}) = 0$ if no trade occurs at time u.

2.2. Utility-based approach to option pricing and hedging

Choosing the utility function $U : \mathbb{R} \to \mathbb{R}$ to be concave and increasing (so it is a risk-averse utility function) and assuming that the investor's goal is to maximize the expected utility of terminal wealth, Hodges and Neuberger (1989) formulate the option hedging problem in terms of the value functions

$$V^i(t,S,x,y) = \sup_{L,M} \mathrm{E}[U(\Omega_T^i) | S_t = S, x_t = x, y_t = y] \tag{12}$$

associated with no call option ($i = 0$), short call ($i = s$) and long call ($i = b$). Denoting the optimal trading strategy for (12) by x_t^i ($i = 0, s, b$), the optimal hedging strategies for the option writer and buyer are given by $x_t^s - x_t^0$ and $x_t^b - x_t^0$, respectively. In particular, under the "risk-neutral" assumption that $\alpha = r$, it can be shown that $x_t^0 \equiv 0$ (i.e., without the option liability, the investor would choose not to invest in the stock at all). Arguing by indifference pricing, the *reservation selling* (resp. *buying*) *price* is defined as the amount of cash P^s (resp. P^b) required *initially* to provide the same expected utility as not selling (resp. buying) the option. Therefore, P^s and P^b satisfy

$$V^s(0,S,x,y + P^s(S,x)) = V^0(0,S,x,y) = V^b(0,S,x,y - P^b(S,x)). \tag{13}$$

Note that Davis (1997) has proposed an alternative utility-based approach to option pricing, which requires that an infinitesimal diversion of funds into the purchase or sale of an option has a neutral effect on the investor's maximum utility. This alternative approach has enabled Monoyios (2003, 2004) to develop a fast numerical algorithm to solve for the associated option hedging strategy.

In particular, if (12) uses the negative exponential utility function $U(z) = 1 - e^{-\gamma z}$, which has constant absolute risk aversion (CARA) γ, then by (8)

$$V^i(t,S,x,y) = 1 - \exp\{-\gamma y e^{r(T-t)}\} H^i(t,S,x), \tag{14}$$

where

$$H^i(t,S,x) = \inf_{L,M} \mathrm{E}[\exp\{\gamma C^i(L,M;t,T)\} | S_t = S, x_t = x] \tag{15}$$

with $H^i(T,S,x) = \exp\{-\gamma Z^i(S,x)\}$. From (14), it follows that $H^i(t,S,x) = 1 - V^i(t,S,x,0)$. Moreover, combining (13) with (14) yields

$$P^s(S,x) = \gamma^{-1} e^{-rT} \log[H^s(0,S,x)/H^0(0,S,x)],$$
$$P^b(S,x) = -\gamma^{-1} e^{-rT} \log[H^b(0,S,x)/H^0(0,S,x)]. \tag{16}$$

In the limit as $\gamma \to \infty$ (the investor is strongly risk averse), Bouchard et al. (2001) have shown that the reservation price (16) approaches the super-replication price of the option plus the liquidation value of the initial endowment.

2.3. Minimization of expected hedging cost under a risk constraint

To begin with, consider the hedging-cost minimization problem, formulated in discrete time and discrete state, of Bensaid et al. (1992). The discussion in the second paragraph of Section 2.1 suggests the following extension of the problem to continuous time and continuous state:

$$V_i(t,S,x) = \inf_{L,M} \mathrm{E}[C^i(L,M;t,T)|S_t = S, x_t = x], \quad i = s, b. \tag{17}$$

However, direct minimization of the expected hedging cost has been shown not to yield a meaningful hedging strategy in continuous time. The optimal strategy in (17) under the risk-neutral measure does not require the investor to do anything before the expiration date, which is similar in nature to the trivial strategy obtained by Soner et al. (1995) for super-replicating an option. This is tantamount to leaving the investor's option position unhedged, and the apparent failure of (17) to produce a non-trivial hedge is due to the fact that the formulation does not address the riskiness of the hedging portfolio. Note that (17) only considers the mean of the random variable $C^i(L,M;t,T)$ and ignores its variance, and that the variability of terminal wealth is not captured in a linear utility function but is reflected well by a CARA utility function that is strictly concave. An example is given in Section 4.3 in which the mean of the hedging error of the optimal strategy in (17) is 0.56, but the standard deviation of the hedging error is 21.5. A reasonable modification of (17) is to consider besides the mean also the variance of the hedging cost. Applying a Lagrange multiplier approach to the problem of minimizing the expected hedging cost subject to a constraint on the variance, which can be regarded as a measure of risk (uncertainty), leads to the following value function whose connection to (14) will be discussed in Section 2.5: For $i = s, b$,

$$\tilde{V}^i(t,S,x,y) = \inf_{L,M} \{\mathrm{E}[C^i(L,M;t,T)|S_t = S, x_t = x, y_t = y] + \omega \,\mathrm{Var}[C^i(L,M;t,T)|S_t = S, x_t = x, y_t = y]\}. \tag{18}$$

Analogous ideas have been used in Markowitz's mean-variance theory of portfolio optimization and in the studies of Clewlow and Hodges (1997) and Zakamouline (2006) that make use of the mean and standard deviation of the hedging error to evaluate the performance of competing hedging strategies. However, the stochastic control problem associated with (18) is prohibitively difficult. We therefore consider in the next two sections simpler alternatives to (18) that attempt to capture its essence.

2.4. Cost-constrained minimization of pathwise risk

As pointed out by Grinold and Kahn (2000, pp. 41–46), "risk is an abstract concept" and other risk measures that depend on the specific investment applications may be more appropriate than variance, which is nevertheless most often used. Motivated by the Black–Scholes theory, Lai and Wong (2004) have introduced a more tractable risk measure which we can use as an alternative to the variance of the hedging cost. To fix ideas, consider a short position in the option. At time $t \in [0,T)$, the option writer's hedging portfolio consists of the option multiplied by -1 and x_t shares of stock. If the market price of the call option is $P(t,S_t)$, with $P(T,S) = (S-K)^+$, the value of the hedging portfolio is $\Pi(t,S_t,x_t) = -P(t,S_t) + x_t S_t$. By (1) and Ito's formula, Π satisfies

$$d\Pi(t,S_t,x_t) = \mu_\Pi(t,S_t,x_t)\,dt + \sigma_\Pi(t,S_t,x_t)\,dW_t, \tag{19}$$

where $\mu_\Pi(t,S_t,x_t)$ is the mean return of the portfolio,

$$\sigma_\Pi^2(t,S,x) = \sigma^2 S^2 [x - \Delta(t,S)]^2 \tag{20}$$

and $\Delta(t,S) = \partial P(t,S)/\partial S$. The instantaneous conditional variance of the hedging portfolio is $\sigma_\Pi^2(t,S_t,x_t)\,dt$. In the absence of transaction costs, Black and Scholes (1973) were able to make $\sigma_\Pi^2(t,S_t,x_t)$ identically zero for all $t \in [0,T)$ and *for every price path* by continuously rebalancing to $x_t = \Delta(t,S_t)$ shares of stock. When rebalancing can only be carried out periodically, the investor has to tolerate positive *pathwise risk*, defined by Lai and Wong (2004) as

$$R(L,M;t,T) = \sigma^2 \int_t^T S_u^2 [x_u - \Delta(u,S_u)]^2\,du. \tag{21}$$

In particular, in the presence of transaction costs, it is natural to ask how the rebalancing times can be chosen to minimize (21) subject to a prescribed upper bound on the hedging cost. A Lagrange multiplier approach leads to the problem

of minimizing

$$J^s(L,M;t,S,x) = \mathrm{E}[\theta R(L,M;t,T) + C^s(L,M;t,T)|S_t = S, x_t = x], \tag{22}$$

where $1/\theta$ is the Lagrange multiplier. Note that instead of cost-constrained minimization of risk, (22) can also be associated with the dual problem of risk-constrained minimization of hedging cost (with θ as the Lagrange multiplier). It turns out that the problem of minimizing (22), which replaces the variance in (18) as a measure of risk by the mean of the pathwise risk (21), is a tractable stochastic control problem whose connection to (18) will be discussed in the next section.

For a long position in the option, the corresponding hedge at time t consists of $+1$ option and x_t shares of stock, so we replace $-\Delta(\cdot,\cdot)$ in (19)–(21) by $+\Delta(\cdot,\cdot)$ and define $J^b(L,M;t,S,x)$ as in (22) but with C^s replaced by C^b. Thus, the value functions of the cost-constrained risk minimization problems for the short and long positions are the following: For $(t,S,x) \in [0,T] \times (0,\infty) \times \mathbb{R}$,

$$V^i(t,S,x) = \inf_{L,M} J^i(L,M;t,S,x), \quad i = s, b \tag{23}$$

subject to the terminal conditions

$$V^i(T,S,x) = -Z^i(S,x), \quad i = s, b. \tag{24}$$

The optimal hedging strategy corresponds to the pair (L,M) that attains $V^i(t,S,x)$ in (23). In particular, we recover the Black–Scholes delta-hedging strategy (i.e., $x_t = \Delta(t,S_t)$ for $i = s$ or $x_t = -\Delta(t,S_t)$ for $i = b$) when $\lambda = \mu = 0$. Formulating option hedging as cost-constrained minimization of pathwise risk (or equivalently, risk-constrained minimization of hedging cost) in (23) provides a natural extension of the Black–Scholes delta-hedging strategy to self-financing strategies that incorporate transaction costs to perform "delta tolerance" hedging. Moreover, the solution to the tractable problem of cost-constrained minimization of risk, which is equivalent to an optimal stopping problem, has provided key insights into the development of a "layered" backward induction algorithm (associated with optimal stopping) in Section 3.2 for solving the utility maximization problem.

2.5. Connections of (14) or (23) to (18): choice of γ or θ

For the stochastic control problem associated with (18), not only does the control (L,M) appear in the objective function in a complicated way, but the objective function also involves the variance of $C^i(L,M;t,T)$. Since the law of iterated conditional expectations $\mathrm{E}(\cdot) = \mathrm{E}[\mathrm{E}(\cdot|\mathscr{F})]$, a key tool in stochastic control, does not have a similar counterpart for $\mathrm{Var}(\cdot)$, we shall convert (18) into a family of stochastic control problems involving the second moment instead of the variance so that the solution to (18) can be obtained by minimizing the value functions of this family.

For notational simplicity, denote $\mathrm{E}(\cdot|S_t = S, x_t = x, y_t = y)$ simply by $\mathrm{E}(\cdot)$ and $C^i(L,M;t,T)$ by C. Let $(\bar{L},\bar{M})$ be the optimal control policy in (18) and denote $C^i(\bar{L},\bar{M};t,T)$ by $\bar{C}$. Let $\eta = 1 - 2\omega\mathrm{E}(\bar{C})$. Since $\mathrm{Var}(C) = \mathrm{E}C^2 - (\mathrm{E}C)^2$,

$$\begin{aligned}
0 &\le \{\mathrm{E}C + \omega\mathrm{E}C^2 - \omega(\mathrm{E}C)^2\} - \{\mathrm{E}\bar{C} + \omega\mathrm{E}\bar{C}^2 - \omega(\mathrm{E}\bar{C})^2\} \\
&= \eta(\mathrm{E}C - \mathrm{E}\bar{C}) + \omega(\mathrm{E}C^2 - \mathrm{E}\bar{C}^2) - \omega(\mathrm{E}C - \mathrm{E}\bar{C})^2 \\
&\le \eta(\mathrm{E}C - \mathrm{E}\bar{C}) + \omega(\mathrm{E}C^2 - \mathrm{E}\bar{C}^2) \\
&= (\eta\mathrm{E}C + \omega\mathrm{E}C^2) - (\eta\mathrm{E}\bar{C} + \omega\mathrm{E}\bar{C}^2).
\end{aligned}$$

Hence the optimal control policy in (18) is also optimal for

$$V_\eta = \inf_{L,M} \mathrm{E}(\eta C + \omega C^2). \tag{25}$$

Although $\mathrm{E}(\bar{C})$ and therefore also $\eta = 1 - 2\omega\mathrm{E}(\bar{C})$ are unknown without solving (18), one can bypass (18) by solving (25) for each given η to obtain the optimal control law (L_η, M_η) and then finding the η that minimizes

$$\mathrm{E}\{C^i(L_\eta, M_\eta; t, T)\} + \omega\,\mathrm{Var}\{C^i(L_\eta, M_\eta; t, T)\}.$$

This converts the "non-standard" stochastic control problem (18) into a family of stochastic control problems of the standard (expectation) form.

With $\gamma = \omega/\eta$, we can reparameterize the family of stochastic control problems (25) as $\inf_{L,M}\mathrm{E}(C + \gamma C^2)$. Instead of γC^2, we can consider more general $g_\gamma(C)$, where g_γ is a strictly convex function for every $\gamma > 0$. This includes as a special case the function

$$g_\gamma(x) = \gamma^{-1}(e^{\gamma x} - 1 - \gamma x) \tag{26}$$

used in the CARA utility maximization problem that can be expressed as

$$V^i(t,S,x,y) = 1 - \gamma\exp\{-\gamma y e^{r(T-t)}\}\left\{\gamma^{-1} + \inf_{L,M}\mathrm{E}[C^i + g_\gamma(C^i)|S_t = S, x_t = x]\right\},$$

where $C^i = C^i(L,M;t,T)$, see (14)–(15). An important computational advantage of CARA utility is that it removes y_t from the stochastic control problem, making it much more tractable, as will be elaborated in Section 3.2. The parameter γ measures the extent to which the loss function deviates from linearity.

We have pointed out in Section 2.3 that under the risk-neutral measure, direct minimization of the hedging cost (17) yields the trivial strategy of not incurring any transaction cost until the expiration date. Since the mean of $C^i(L,M;t,T)$ ignores the variability of the random terminal payoff $Z^i(S_T,x_T)$, we include the non-linear term $g_\gamma(C^i(L,M;t,T))$ to bring indirectly such variability into the stochastic control problem $\inf_{L,M}\mathrm{E}\{C+g_\gamma(C)\}$. Using (26) instead of $g_\gamma(x)=\gamma x^2$, which is directly related to (18) and which can be regarded as a Taylor approximation to (26), makes the stochastic control problem much more tractable. Instead of $g_\gamma(C)$, the basic idea in Section 2.4 is to use a more general random variable θR so that the control problem is tractable and has an optimal rule that periodically buys and sells stock, instead of postponing to the expiration date that would incur risk. One such choice of R is the Black–Scholes risk measure (21). With a suitably chosen parameter θ in $C+\theta R$, we shall show in Section 4.3 that the control policy which minimizes $\mathrm{E}(C+\theta R)$ can achieve performance comparable to that of $\mathrm{E}\{C+g_\gamma(C)\}$ for suitably chosen γ. This approach of choosing an optimal parameter (θ or γ) is similar in spirit to that of Mohamed (1994), who selects the "best" parameter for each of a number of option hedging strategies by minimizing the "95% risk of loss."

The performance criterion we use in Section 4.3 is $\mathrm{E}C^2-a\mathrm{E}C$ for some prescribed constant $a>0$, which corresponds to a special case of (25) and which is motivated by the root mean squared hedging error introduced by Hutchinson et al. (1994). We propose to use utility maximization (14) or cost-constrained risk minimization (23) as a tractable approximation to the stochastic control problem of minimizing $\mathrm{E}(C^2-aC)$. Specifically, with a given risk-aversion coefficient γ belonging to a grid of values, we solve the utility maximization problem using the algorithm described in Section 3.2 for the optimal control policy $(\tilde{L}_\gamma,\tilde{M}_\gamma)$ and then choose the $(\tilde{L}_\gamma,\tilde{M}_\gamma)$ that minimizes $\mathrm{E}(C^2-aC)$. Likewise, for the cost-constrained risk minimization problem, we solve for the optimal control policy (L^*_θ,M^*_θ) using the algorithm described in Section 3.1 for every θ over a grid and then choose the (L^*_θ,M^*_θ) that minimizes $\mathrm{E}(C^2-aC)$.

3. Computational algorithms for singular stochastic control

3.1. Equivalence of (23) to optimal stopping and an associated algorithm

In view of (6), (21) and (22), the value functions $V^i(t,S,x)$ $(i=s,b)$ in (23) have the following form:

$$V(t,S,x)=\inf_{L,M}\mathrm{E}\left[\int_t^T F(u,S_u,x_u)\,du+\Psi(L,M;t,T)+g(S_T,x_T)|S_t=S,x_t=x\right] \tag{27}$$

with $F(u,S,x)=\theta\sigma^2S^2[x-\Delta(u,S)]^2$ for a short call ($i=s$) or $\theta\sigma^2S^2[x+\Delta(u,S)]^2$ for a long call ($i=b$), and $g(S,x)=-Z^s(S,x)$ or $-Z^b(S,x)$, respectively. It can be shown that the state space $[0,T]\times(0,\infty)\times\mathbb{R}$ is partitioned into (i) a "buy stock" region $\mathscr{B}$, (ii) a "sell stock" region $\mathscr{S}$ and (iii) a "no-transaction" region $\mathscr{N}$, respectively. With the change of variables

$$\beta=\alpha/\sigma^2,\quad \rho=r/\sigma^2,\quad s=\sigma^2(t-T),\quad z=\log(S/K)-(\beta-1/2)s, \tag{28}$$

we find that $v(s,z,x)=V(t,S,x)/K$ is the value function of the following singular stochastic control problem for the Brownian motion $\{Z_u,\ u\le 0\}$:

$$v(s,z,x)=\inf_{L,M}\mathrm{E}\left[\int_s^0 f(u,Z_u,x_u)\,du+\psi(L,M;s)+\tilde{g}(Z_0,x_0)|Z_s=z,x_s=x\right], \tag{29}$$

where $f(s,z,x)=F(t,S,x)/K\sigma^2$, $\psi(L,M;s)=\int_{[s,0)}e^{Z_u+(\beta-\rho-1/2)u}(a\,dL_u-b\,dM_u)$ and $\tilde{g}(z,x)=g(Ke^z,x)/K$. As shown in Lai et al. (2009), the function

$$w(s,z,x)=\frac{\partial v}{\partial x}(s,z,x),\quad (s,z,x)\in[-\sigma^2T,0]\times\mathbb{R}\times\mathbb{R}, \tag{30}$$

is the value function of the optimal stopping problem associated with the following Dynkin game:

$$\begin{aligned} w(s,z,x)&=\underline{w}(s,z,x):=\sup_{\tau_L\in\mathscr{T}(s,0)}\inf_{\tau_M\in\mathscr{T}(s,0)}I(\tau_L,\tau_M;s,z,x)\\ &=\overline{w}(s,z,x):=\inf_{\tau_M\in\mathscr{T}(s,0)}\sup_{\tau_L\in\mathscr{T}(s,0)}I(\tau_L,\tau_M;s,z,x), \end{aligned} \tag{31}$$

where $\mathscr{T}(a,b)$ denotes the set of stopping times taking values between a and b $(>a)$, $f_x(s,z,x)=\partial f(s,z,x)/\partial x$ and $\tilde{g}_x(z,x)=\partial\tilde{g}(z,x)/\partial x$ are non-decreasing in x (see (37)), and

$$\begin{aligned} &I(\tau_1,\tau_2;s,z,x)\\ &=\mathrm{E}\left[\int_s^{\tau_1\wedge\tau_2}f_x(u,Z_u,x_u)\,du+G(\tau_1,\tau_2,Z_{\tau_1},Z_{\tau_2})+\tilde{g}_x(Z_0,x_0)\mathbb{I}_{\{\tau_1=\tau_2=0\}}|Z_s=z,x_s=x\right],\\ &G(s_1,s_2,z_1,z_2)=\begin{cases}-ae^{z_1+(\beta-\rho-1/2)s_1} & \text{if } s_1<s_2<0,\\ -be^{z_2+(\beta-\rho-1/2)s_2} & \text{if } s_2<s_1<0.\end{cases} \end{aligned} \tag{32}$$

In addition to relationship (30) between the value functions v and w, the optimal continuation region of the Dynkin game (31) coincides with the no-transaction region of the singular stochastic control problem (29) in the following sense: If

(L^*, M^*) is the optimal control of (29) and $\tau_L^* = \inf\{u \in [s,0) : L_u^* > 0\}$, $\tau_M^* = \inf\{u \in [s,0) : M_u^* > 0\}$ $(\inf \emptyset = 0)$, then $w(s,z,x) = I(\tau_L^*, \tau_M^*; s, z, x)$. Moreover, letting

$$w_b(s,z,x) = -ae^{z+(\beta-\rho-1/2)s} \quad \text{and} \quad w_s(s,z,x) = -be^{z+(\beta-\rho-1/2)s}, \tag{33}$$

w solves the free boundary problem associated with optimal stopping in the Dynkin game:

$$\frac{\partial w}{\partial s} + \frac{1}{2}\frac{\partial^2 w}{\partial z^2} + f_x = 0 \quad \text{in } \mathscr{N}, \tag{34a}$$
$$w(s,z,x) = w_b(s,z,x) \quad \text{in } \mathscr{B}, \tag{34b}$$
$$w(s,z,x) = w_s(s,z,x) \quad \text{in } \mathscr{S}, \tag{34c}$$
$$w(0,z,x) = \tilde{g}_x(z,x). \tag{34d}$$

Since $w(s,z,x)$ is non-decreasing in x, the region $\mathscr{B}$ in (34b) has an upper boundary $X_b(s,z)$ which is the largest x for which $w(s,z,x) = w_b(s,z,x)$; this is the *buy boundary*. Similarly, the region $\mathscr{S}$ in (34c) has a lower boundary $X_s(s,z)$ which is the smallest x for which $w(s,z,x) = w_s(s,z,x)$; this is the *sell boundary*. Recall that $a \geq 1 \geq b > 0$.

In view of the functional central limit theorem, we can approximate standard Brownian motion by a symmetric Bernoulli random walk. Since the horizon for problem (31) is always 0, only one numerical program for each set of parameters (β, ρ) need be implemented for all expiration dates T. Also, from (28), $s = -\sigma^2 T$ at time $t = 0$. Therefore, letting $T_{\max}$ denote the largest expiration date of interest and taking small positive δ and ε such that $N := \sigma^2 T_{\max}/\delta$ is an integer, Lai et al. (2009) have introduced the following backward induction to solve optimal stopping problems of the type (31). For $i = 1, 2, \ldots, N$,

$$w(s_i,z,x) = \begin{cases} w_b(s_i,z,x) & \text{if } \widetilde{w}(s_i,z,x) < w_b(s_i,z,x), \\ w_s(s_i,z,x) & \text{if } \widetilde{w}(s_i,z,x) > w_s(s_i,z,x), \\ \widetilde{w}(s_i,z,x) & \text{otherwise}, \end{cases} \tag{35}$$

where $s_0 = 0$, $s_i = s_{i-1} - \delta$, $z \in \mathbf{Z}_\delta = \{0, \pm\sqrt{\delta}, \pm 2\sqrt{\delta}, \ldots\}$, $x \in \mathbf{X}_\varepsilon = \{0, \pm\varepsilon, \pm 2\varepsilon, \ldots\}$ and

$$\widetilde{w}(s,z,x) = \delta f_x(s,z,x) + [w(s+\delta, z+\sqrt{\delta}, x) + w(s+\delta, z-\sqrt{\delta}, x)]/2. \tag{36}$$

This suggests the backward induction algorithm described below to solve for $X_b(s_i,z)$ as the largest x for which $\widetilde{w}(s_i,z,x) \leq w_b(s_i,z,x)$ and for $X_s(s_i,z)$ as the smallest x for which $\widetilde{w}(s_i,z,x) \geq w_s(s_i,z,x)$.

Algorithm 1. Backward induction Algorithm 1.

For $i = 1, 2, \ldots, N$ and $z \in \mathbf{Z}_\delta$:

(i) Starting at $x_0 \in \mathbf{X}_\varepsilon$ with $\widetilde{w}(s_i,z,x_0) < w_b(s_i,z,x_0)$, search for the first $j \in \{1,2,\ldots\}$ (denoted by j^*) for which $\widetilde{w}(s_i,z,x_0+j\varepsilon) \geq w_b(s_i,z,x_0+j\varepsilon)$ and set $X_b(s_i,z) = x_0 + j^*\varepsilon$.

(ii) For $j \in \{1,2,\ldots\}$, let $x_j = X_b(s_i,z) + j\varepsilon$. Compute, and store for use at s_{i+1}, $w(s_i,z,x_j) = \widetilde{w}(s_i,z,x_j)$ as defined by (36). Search for the first j (denoted by j^*) for which $\widetilde{w}(s_i,z,x_j) \geq w_s(s_i,z,x_j)$ and set $X_s(s_i,z) = X_b(s_i,z) + j^*\varepsilon$.

(iii) For $x \in \mathbf{X}_\varepsilon$ outside the interval $[X_b(s_i,z), X_s(s_i,z)]$, set $w(s_i,z,x) = w_b(s_i,z,x)$ or $w_s(s_i,z,x)$ as defined by (33) according to whether $x \leq X_b(s_i,z)$ or $x \geq X_s(s_i,z)$.

For the cost-constrained risk minimization problem (23), the transformations (28) lead to (29) with

$$f(s,z,x) = \begin{cases} \theta K e^{2z+(2\beta-1)s}[x - D(s,z)]^2 & \text{for short call } (i = s), \\ \theta K e^{2z+(2\beta-1)s}[x + D(s,z)]^2 & \text{for long call } (i = b), \end{cases}$$
$$v(0,z,x) = -A^i(z,x), \quad i = s, b,$$

where $D(s,z) = \Delta(t,S)$, and, corresponding to definitions (3)–(5) of terminal settlement value,

$$A^0(z,x) = xe^z(a\mathbb{I}_{\{x<0\}} + b\mathbb{I}_{\{x\geq 0\}}),$$
$$A^i(z,x) = A^0(z, x - D^i(z)) + D^i(z) \text{ for asset settlement}, \quad i = s, b,$$
$$A^i(z,x) = A^0(z,x) - (e^z - 1)D^i(z) \text{ for cash settlement}, \quad i = s, b,$$

with $D^s(z) = \mathbb{I}_{\{z>0\}}$ (short call) and $D^b(z) = -\mathbb{I}_{\{z>0\}}$ (long call). Thus, Algorithm 1 uses

$$f_x(s,z,x) = \begin{cases} 2\theta K e^{2z+(2\beta-1)s}[x - D(s,z)] & \text{for short call } (i = s), \\ 2\theta K e^{2z+(2\beta-1)s}[x + D(s,z)] & \text{for long call } (i = b), \end{cases} \tag{37}$$
$$w(0,z,x) = -B^i(z,x), \quad i = s, b,$$

where $B^i(z,x) := \partial A^i(z,x)/\partial x$ is given by $B^0(z,x) = A^0(z,x)/x$ and, for $i = s, b$, $B^i(z,x) = B^0(z, x - D^i(z))$ under asset settlement and $B^i(z,x) = B^0(z,x)$ under cash settlement. Note that θK can be treated as a single parameter which controls the width of the no-transaction region.

3.2. A coupled backward induction algorithm for (12) for CARA utility

From the Hamilton–Jacobi–Bellman equation of dynamic programming in continuous time, it follows that $V^i(t,S,x,y)$ given by (12) satisfies the free boundary problem (FBP)

$$\begin{aligned}
&\frac{\partial V^i}{\partial t} + \alpha S \frac{\partial V^i}{\partial S} + \frac{\sigma^2 S^2}{2}\frac{\partial^2 V^i}{\partial S^2} + ry\frac{\partial V^i}{\partial y} = 0 \quad \text{in } \mathscr{N},\\
&\frac{\partial V^i}{\partial x}(t,S,x,y) = aS\frac{\partial V^i}{\partial y}(t,S,x,y) \quad \text{in } \mathscr{B},\\
&\frac{\partial V^i}{\partial x}(t,S,x,y) = bS\frac{\partial V^i}{\partial y}(t,S,x,y) \quad \text{in } \mathscr{S},\\
&V^i(T,S,x,y) = U(y + Z^i(S,x));
\end{aligned}$$

see Davis et al. (1993) who also show that for the CARA utility function $U(z) = 1 - e^{-\gamma z}$, $H^i(t,S,x)$ given by (15) satisfies the FBP

$$\begin{aligned}
&\frac{\partial H^i}{\partial t} + \alpha S \frac{\partial H^i}{\partial S} + \frac{\sigma^2 S^2}{2}\frac{\partial^2 H^i}{\partial S^2} = 0, \quad x \in [X_b(t,S), X_s(t,S)],\\
&\frac{\partial H^i}{\partial x}(t,S,x) = -a\gamma S e^{r(T-t)} H^i(t,S,x), \quad x \le X_b(t,S),\\
&\frac{\partial H^i}{\partial x}(t,S,x) = -b\gamma S e^{r(T-t)} H^i(t,S,x), \quad x \ge X_s(t,S),\\
&H^i(T,S,x) = \exp\{-\gamma Z^i(S,x)\}.
\end{aligned}$$

Applying the change of variables (28) and letting $h^i(s,z,x) = H^i(t,S,x)$, this FBP can be rewritten as

$$\frac{\partial h^i}{\partial s} + \frac{1}{2}\frac{\partial^2 h^i}{\partial z^2} = 0, \quad x \in [X_b(s,z), X_s(s,z)], \tag{38a}$$

$$\frac{\partial h^i}{\partial x}(s,z,x) = -a\gamma K e^{z+(\beta-\rho-1/2)s} h^i(s,z,x), \quad x \le X_b(s,z), \tag{38b}$$

$$\frac{\partial h^i}{\partial x}(s,z,x) = -b\gamma K e^{z+(\beta-\rho-1/2)s} h^i(s,z,x), \quad x \ge X_s(s,z), \tag{38c}$$

$$h^i(0,z,x) = \exp\{-\gamma K A^i(z,x)\}. \tag{38d}$$

Since $\partial/\partial s + (1/2)\partial^2/\partial z^2$ is the infinitesimal generator of space-time Brownian motion, this means that while (s, Z_s) is inside the no-transaction region, the dynamics of $h^i(s, Z_s, x_s)$ is driven by the standard Brownian motion $\{Z_s,\ s \le 0\}$. In the buy and sell regions, it follows from (38b) and (38c) that

$$h^i(s,z,x) = \exp\{-a\gamma K e^{z+(\beta-\rho-1/2)s}[x - X_b(s,z)]\} h^i(s,z,X_b(s,z)), \quad x \le X_b(s,z), \tag{39a}$$

$$h^i(s,z,x) = \exp\{-b\gamma K e^{z+(\beta-\rho-1/2)s}[x - X_s(s,z)]\} h^i(s,z,X_s(s,z)), \quad x \ge X_s(s,z). \tag{39b}$$

Finally, if we let $w^i = \partial h^i/\partial x$, then w^i satisfies the FBP

$$\frac{\partial w^i}{\partial s} + \frac{1}{2}\frac{\partial^2 w^i}{\partial z^2} = 0, \quad x \in [X_b(s,z), X_s(s,z)], \tag{40a}$$

$$w^i(s,z,x) = -a\gamma K e^{z+(\beta-\rho-1/2)s} h^i(s,z,x), \quad x \le X_b(s,z), \tag{40b}$$

$$w^i(s,z,x) = -b\gamma K e^{z+(\beta-\rho-1/2)s} h^i(s,z,x), \quad x \ge X_s(s,z), \tag{40c}$$

$$w^i(0,z,x) = -\gamma K B^i(z,x) h^i(0,z,x). \tag{40d}$$

In the sequel, we fix $i = 0, s, b$ and drop the superscript i in w^i and h^i for notational simplicity. If the function $h(s,z,x)$ is known, then by analogy to (34), the FBP (40) is an optimal stopping problem associated with a Dynkin game, and its solution can be computed by backward induction (35), with

$$w_b(s,z,x) = -a\gamma K e^{z+(\beta-\rho-1/2)s} h(s,z,x), \tag{41a}$$

$$w_s(s,z,x) = -b\gamma K e^{z+(\beta-\rho-1/2)s} h(s,z,x) \tag{41b}$$

and $f_x \equiv 0$ in (36). As in Section 3.1, $X_b(s_i,z)$ is the largest x for which $\widetilde{w}(s_i,z,x) \le w_b(s_i,z,x)$ and $X_s(s_i,z)$ is the smallest x for which $\widetilde{w}(s_i,z,x) \ge w_s(s_i,z,x)$. On the other hand, if the boundaries $X_b(s,z)$ and $X_s(s,z)$ are given, then problem (38) can also

be solved by backward induction: For $z \in \mathbf{Z}_\delta$, define $h(s_i, z, x)$ by (39) (with s replaced by s_i) if $x \in \mathbf{X}_\varepsilon$ is outside the interval $[X_b(s_i, z), X_s(s_i, z)]$, and let $h(s_i, z, x) = \widetilde{h}(s_i, z, x)$ with

$$\widetilde{h}(s, z, x) = [h(s+\delta, z+\sqrt{\delta}, x) + h(s+\delta, z-\sqrt{\delta}, x)]/2 \tag{42}$$

if $x \in \mathbf{X}_\varepsilon \cap [X_b(s_i, z), X_s(s_i, z)]$. Since (38) can be equivalently expressed as

$$\min\{\partial h/\partial x + a\gamma K e^{z+(\beta+\rho-1/2)s} h, -(\partial h/\partial x + b\gamma K e^{z+(\beta-\rho-1/2)s} h), \partial h/\partial s + (1/2)\partial^2 h/\partial z^2\} = 0,$$

it follows that $\partial h/\partial s + (1/2)\partial^2 h/\partial z^2 > 0$ for $x \notin [X_b(s, z), X_s(s, z)]$. Using an argument based on the finite difference method, it can be shown that $h(s, z, x) < \widetilde{h}(s, z, x)$ in the buy and sell regions; see Lai et al. (2009). Defining $\widetilde{w}_b$ and $\widetilde{w}_s$ as in (41a) and (41b) but with h replaced by $\widetilde{h}$, this suggests the coupled backward induction algorithm described below to solve for $X_b(s_i, z)$ and $X_s(s_i, z)$, as well as to compute values of $h(s_i, z, x)$ for $x \in \mathbf{X}_\varepsilon \cap [X_b(s_i, z), X_s(s_i, z)]$.

Algorithm 2. Backward induction Algorithm 2.

For $i = 1, 2, \ldots, N$ and
$z \in \mathbf{Z}_\delta$:

(i) Starting at $x_0 \in \mathbf{X}_\varepsilon$ with $\widetilde{w}(s_i, z, x_0) < \widetilde{w}_b(s_i, z, x_0)$, search for the first $j \in \{1, 2, \ldots\}$(denoted by j^*) for which $\widetilde{w}(s_i, z, x_0 + j\varepsilon) \geq \widetilde{w}_b(s_i, z, x_0 + j\varepsilon)$ and set $X_b(s_i, z) = x_i + j^*\varepsilon$.

(ii) For $j \in \{1, 2, \ldots\}$, let $x_j = X_b(s_i, z) + j\varepsilon$. Compute, and store for use at s_{i+1}, $w(s_i, z, x_j) = \widetilde{w}(s_i, z, x_j)$ as defined by (36) with $f_x \equiv 0$ and $h(s_i, z, x_j) = \widetilde{h}(s_i, z, x_j)$ by (42). Search for the first j (denoted by j^*) for which $\widetilde{w}(s_i, z, x_j) \geq \widetilde{w}_s(s_i, z, x_j)$ and set $X_s(s_i, z) = X_b(s_i, z) + j^*\varepsilon$.

(iii) For $x \in \mathbf{X}_\varepsilon$ outside the interval $[X_b(s_i, z), X_s(s_i, z)]$, compute $h(s_i, z, x)$ using (39) and set $w(s_i, z, x) = w_b(s_i, z, x)$ or $w_s(s_i, z, x)$ as defined by (41) according to whether $x \leq X_b(s_i, z)$ or $x \geq X_s(s_i, z)$.

Whereas the discrete-time dynamic programming algorithms of Davis et al. (1993), Clewlow and Hodges (1997) and Zakamouline (2006) have to perform an additional non-linear maximization to identify the optimal trade size at each time step, the much simpler Algorithm 2 avoids such maximization by solving the coupled problem for (w, h) rather than just for h (or equivalently H). To improve numerical stability in implementing Algorithm 2, we recommend to compute and store values of $(\log(-w), \log h)$, rather than of (w, h) directly. For the convergence of Algorithm 2 and its variants in other applications to singular stochastic control problems that cannot be reduced to optimal stopping problems, see Lai et al. (2009).

4. Numerical results

4.1. The buy–sell boundaries for the utility-maximization approach

As an illustration of Algorithm 2, we consider the hedging of a short call option with $K = 100$, $T = 0.5$ and asset or cash settlement (both in Fig. 1), for the parameter configuration $\alpha = r = 0$, $\sigma = 0.3$, $\lambda = \mu = 0.02$ and $\gamma = 1$, which has also been used by Clewlow and Hodges (1997) in their numerical study. By choosing $\alpha = r$, the optimal hedging strategy for the utility-maximization approach can be characterized by a pair (rather than two pairs) of buy–sell boundaries; see Section 2.2. The boundaries obtained using Algorithm 2 with $\delta = 10^{-4}$ and $\varepsilon = 10^{-5}$ for $S = 90, 100, 110$ are in close agreement with those obtained by Clewlow and Hodges (1997) under cash settlement (their Fig. 5). If settlement must be made in cash, it is optimal to liquidate any excess position in the underlying stock "sooner rather than later," resulting in both boundaries tending to 0 just prior to expiration. In the case of asset settlement as defined by Davis et al. (1993), we find that the boundaries tend to 0 for an out-of-the-money call, or to 1 for an in-the-money call because the option writer needs to deliver a physical share of stock when the option is exercised; see the top panel of Fig. 1.

4.2. The buy–sell boundaries for cost-constrained risk minimization

In order to use the cost-constrained risk-minimization approach for option hedging, one needs to first specify the delta to be used in (20). A common practice uses market data consisting of the option price, its strike price, time to expiration, the underlying stock price and the short-term interest rate to evaluate an implied volatility. If the implied volatility changes relatively slowly as the stock price varies, the Black–Scholes delta evaluated at the implied volatility is a reasonably good approximation of the actual market delta. Asymptotic and numerical analyses of the CARA utility-maximization problem provide further justification for the use of the Black–Scholes delta with a volatility adjustment; see Whalley and Wilmott (1997) and Zakamouline (2006). To compare our results with those of the CARA utility-maximization approach, we set the delta to be the Black–Scholes delta with an adjusted volatility $\hat{\sigma}$, i.e.,

$$\Delta(t, S) = \Delta_{\mathrm{BS}}(t, S; \hat{\sigma}) = \Phi(d_1(t, S; \hat{\sigma})), \tag{43}$$

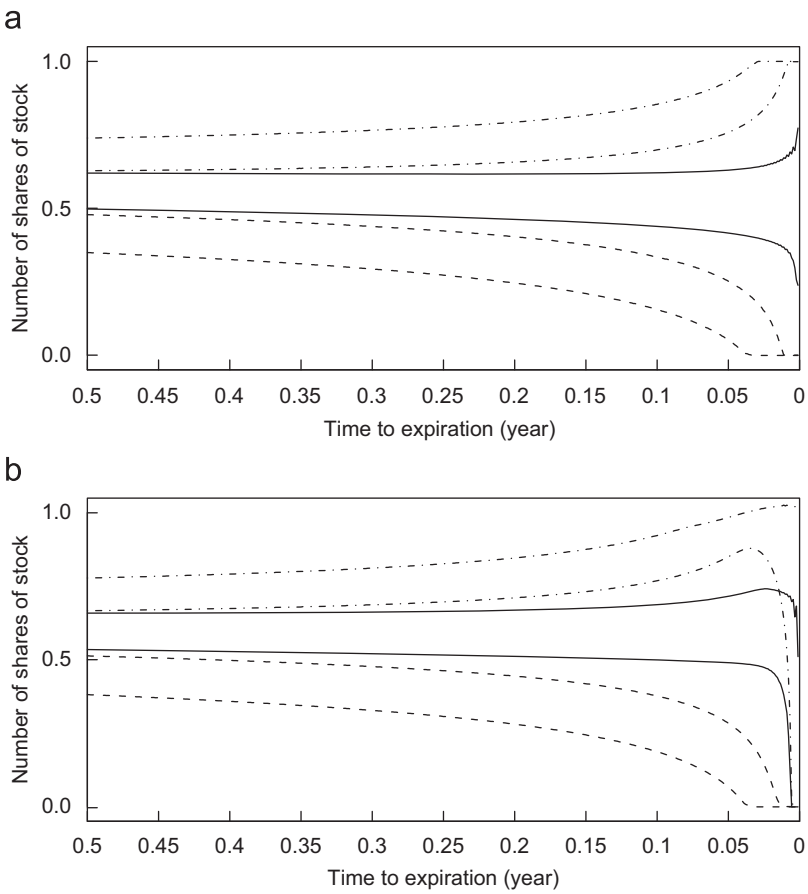

Fig. 1. Optimal buy (lower) and sell (upper) boundaries $X_b(t,S)$ and $X_s(t,S)$ from CARA utility maximization for a short call with $K = 100$ and (a) asset settlement or (b) cash settlement. The boundaries are obtained using Algorithm 2 for $\alpha = r = 0$, $\sigma = 0.3$, $\gamma = 1$, $\lambda = \mu = 0.02$, $0 \le T - t \le 0.5$ and $S = 90$, 100 and 110 (dashed, solid and dot–dash lines, respectively). For each pair of boundaries, the "buy stock" region is below $X_b(t,S)$ and the "sell stock" region is above $X_s(t,S)$; the no-transaction region is between the two boundaries.

where

$$d_1(t,S;\sigma) = [\log(S/K) + (r + \sigma^2/2)(T-t)]/\{\sigma\sqrt{T-t}\}. \tag{44}$$

The optimal buy and sell boundaries for the writer of a call option with $K = 100$ and $T = 0.5$ are shown in Fig. 2 for the same market parameters as those used in Section 4.1, Lagrange multiplier $\theta = 3$ and adjusted volatility $\hat{\sigma} = 1.3244\sigma$, where σ is the volatility parameter of the stock price process (1). The rationale behind this choice of the adjusted volatility is explained in Section 4.4. The boundaries are obtained by first applying Algorithm 1 with $\delta = 10^{-4}$ and $\varepsilon = 10^{-6}$ and then by mapping back to the original (t,S)-scale for $S = 90, 100, 110$. A comparison of Figs. 1 and 2 shows that the boundaries obtained by using the cost-constrained risk-minimization approach have features that are similar to those obtained by using utility maximization.

4.3. A comparative study of the performance of hedging strategies

A simple way to assess the performance of an option hedging strategy in the presence of transaction costs is to consider the mean and the standard deviation of the hedging cost (C^s or C^b). In fact, Clewlow and Hodges (1997) and Zakamouline (2006) have used a similar idea to compare various hedging strategies by plotting the mean versus the standard deviation of the *hedging error*, which is a linear transformation of the hedging cost defined by

$$\text{err} = \begin{cases} P(0,S_0) - e^{-rT}C^s & \text{for a short call,} \\ -P(0,S_0) - e^{-rT}C^b & \text{for a long call,} \end{cases} \tag{45}$$

where $P(t,S)$ denotes the market price of a European call option. A variation of this approach is adopted by Martellini and Priaulet (2002), who plot the standard deviation of the hedging error against the expected transaction costs incurred by the strategy. We shall combine the mean and standard deviation of (45) into a single summary statistic $\eta := \sqrt{\mathrm{E}\{(\text{err})^2\}}$, which

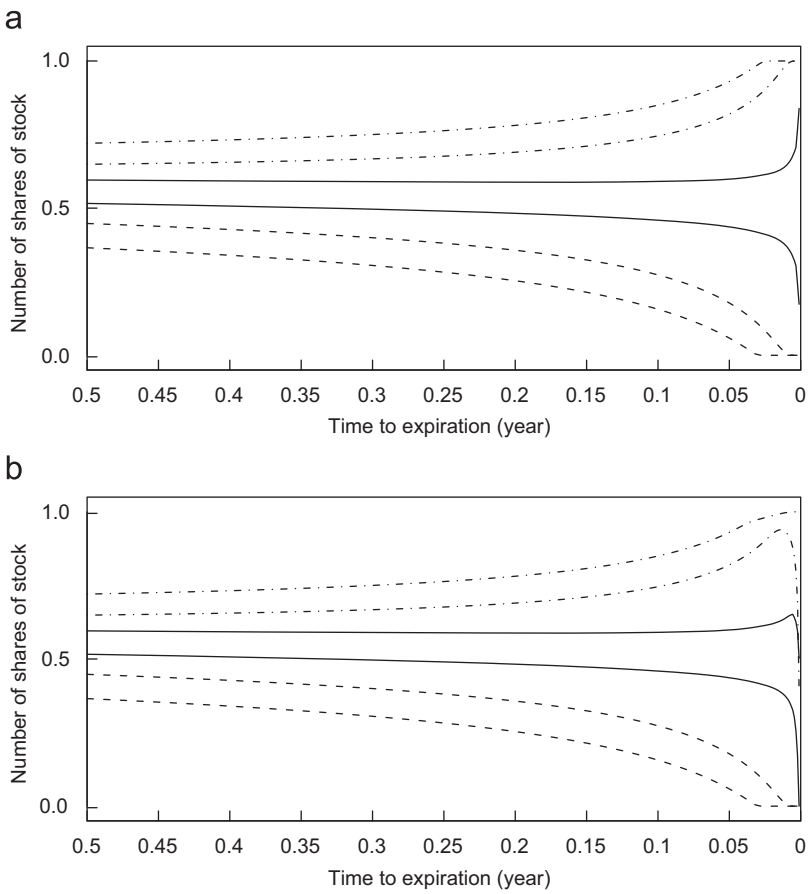

Fig. 2. Optimal buy (lower) and sell (upper) boundaries $X_b(t,S)$ and $X_s(t,S)$ from cost-constrained minimization of pathwise risk for a short call with $K = 100$ and (a) asset settlement or (b) cash settlement. The boundaries are obtained using Algorithm 1 for $\alpha = r = 0$, $\sigma = 0.3$, $\theta = 3$, $\hat{\sigma} = 1.3244\sigma$ (adjusted volatility), $\lambda = \mu = 0.02$, $0 \le T - t \le 0.5$ and $S = 90, 100$ and 110 (dashed, solid and dot–dash lines, respectively). For each pair of boundaries, the "buy stock" region is below $X_b(t,S)$ and the "sell stock" region is above $X_s(t,S)$; the no-transaction region is between the two boundaries.

has been used by Hutchinson et al. (1994) to measure the hedging error in the absence of transaction costs. Besides the mean and standard deviation of the hedging error, we also consider $\kappa := \sqrt{\mathrm{E}\{R(L,M;0,T)\}}$, which Lai and Wong (2004) use to measure the hedging performance in the absence of transaction costs.

We use the following procedure to compute by Monte Carlo simulations the mean and standard deviation (and hence also η) of the hedging error and κ (measure of pathwise risk) for each strategy. For the cost-constrained risk minimization or the CARA utility-maximization strategy, Algorithm 1 or 2 is used to compute the buy–sell boundaries $X_b(s,z)$ and $X_s(s,z)$ for a given Lagrange multiplier θ or risk-aversion coefficient γ. These boundaries are then mapped back to the (t,S)-grid on which the optimal holding of stock is determined as in (9), and C^s or C^b and (21) are computed for each of 100,000 simulated stock price paths. To illustrate how to choose θ or γ appropriately to reflect the tradeoff between the mean and standard deviation of the hedging error, we consider hedging a short call option with $S = K = 100$, $\alpha = r = 0$, $\sigma = 0.3$, $T - t = 1$ (as in Sections 4.1 and 4.2) and $\lambda = \mu = 0.01$, under two market scenarios. In the first scenario, the market price $P(t,S)$ is taken to be the Black–Scholes price of an European call option:

$$p_{\mathrm{BS}}(t,S;\sigma) = S\Phi(d_1(t,S;\sigma)) - Ke^{-r(T-t)}\Phi(d_2(t,S;\sigma)),$$

where d_1 is defined by (44) and $d_2(t,S;\sigma) = d_1(t,S;\sigma) - \sigma\sqrt{T-t}$. In the second scenario, the market price $P(t,S)$ is taken to be the reservation selling price of the European call option obtained under CARA utility with $\gamma = 1.5$. This reservation selling price is equivalent to $p_{\mathrm{BS}}(t,S;\hat{\sigma})$, the Black–Scholes price with adjusted volatility $\hat{\sigma} = 0.375$.

The simulated values of η and κ together with the mean and standard deviation of the hedging error are given in Table 1 under the first market scenario and in Table 2 under the second market scenario for a coarse grid of values of γ or θ. Since η summarizes the tradeoff between the mean and standard deviation of the hedging error, we use it as a guide to identifying the "best" value of γ or θ on the coarse grid presented in Tables 1 and 2. For the second scenario, Table 2 presents that the minimum values of η are 0.9889 and 0.7990, attained at $\gamma = 7$ and at $\theta = 6$ for the utility-maximized strategy and for the

Table 1
Mean and standard deviation (SD) of the hedging error, η and κ for different strategies under the Black–Scholes market price.

CARA utility maximization							
γ	0.02	0.06	0.2	0.6	2	6	20
Mean	1.0080	1.1643	1.4331	1.7526	2.2526	2.8987	3.8948
SD	5.0893	3.4712	2.0838	1.5186	1.1523	0.9969	0.9588
η	5.1882	3.6612	2.5290	*2.3190*	2.5302	3.0654	4.0111
κ	4.8954	3.3486	2.0348	1.5787	1.4657	1.6605	2.0518
Cost-constrained pathwise-risk minimization							
θ	0.01	0.03	0.1	0.3	1	3	10
Mean	1.0417	1.1514	1.4216	1.7380	2.2574	2.9394	4.0017
SD	4.3844	3.5355	2.1400	1.6152	1.3480	1.3959	1.7432
η	4.5065	3.7182	2.5692	*2.3727*	2.6293	3.2540	4.3649
κ	4.2105	3.3738	1.9722	1.3760	0.9199	0.6285	0.4006
Discrete-time Black–Scholes delta hedging							
δt	1/6	1/10	1/20	1/24	1/30	1/40	1/60
Mean	1.4420	1.6415	2.0353	2.1603	2.3448	2.5964	3.0444
SD	4.1143	3.2645	2.4211	2.2508	2.0708	1.8869	1.7269
η	4.3597	3.6540	3.1629	*3.1198*	3.1283	3.2096	3.5001
κ	0	0	0	0	0	0	0
Leland: discrete-time terminal-payoff replication							
δt	1/6	1/10	1/20	1/24	1/30	1/40	1/60
Mean	1.4281	1.6135	1.9732	2.0852	2.2496	2.4714	2.8530
SD	4.0955	3.2321	2.3492	2.1621	1.9465	1.7126	1.4249
η	4.3373	3.6125	3.0680	3.0038	*2.9749*	3.0068	3.1890
κ	0.3002	0.3708	0.4976	0.5375	0.5913	0.6676	0.7910

The best values of γ, θ and δt are shown in italics and chosen on the basis of η.

Table 2
Mean and standard deviation (SD) of the hedging error, η and κ for different strategies under the CARA utility-based (with $\gamma = 1.5$) market price.

CARA utility maximization							
γ	0.07	0.3	0.7	3	7	30	70
Mean	−1.7501	−1.4285	−1.1406	−0.4704	0.0578	1.3868	2.4960
SD	3.1681	1.8939	1.4560	1.0712	0.9872	0.9480	0.9065
η	3.6194	2.3723	1.8495	1.1699	*0.9889*	1.6799	2.6555
κ	2.6375	1.4405	1.0293	0.6988	0.7649	1.2026	1.5171
Cost-constrained pathwise-risk minimization							
θ	0.06	0.2	0.6	2	6	20	60
Mean	−1.6992	−1.4505	−1.1091	−0.5832	0.1448	1.2001	2.5849
SD	2.8353	1.9880	1.4538	1.0425	0.7858	0.8097	1.2227
η	3.3055	2.4609	1.8286	1.1946	*0.7990*	1.4477	2.8595
κ	2.2050	1.4310	0.9837	0.6651	0.4430	0.2807	0.1608
Discrete-time Black–Scholes delta hedging							
δt	1/10	1/20	1/30	1/60	1/72	1/90	1/120
Mean	−1.3081	−0.9143	−0.6048	−0.0948	0.3241	0.6336	1.0923
SD	3.2645	2.4211	2.0708	1.7269	1.6783	1.6679	1.7042
η	3.5168	2.5880	2.1573	1.7295	*1.7093*	1.7842	2.0242
κ	1.0625	1.0387	1.0304	1.0222	1.0206	1.0199	1.0179

Table 2. (*continued*)

Leland: discrete-time terminal-payoff replication							
δt:	1/10	1/20	1/30	1/60	1/72	1/90	1/120
Mean	−1.3361	−0.9764	−0.7000	−0.0966	0.0926	0.3460	0.7143
SD	3.2321	2.3492	1.9465	1.4249	1.3069	1.1834	1.0429
η	3.4974	2.5440	2.0686	1.4282	1.3102	*1.2330*	1.2640
κ	0.6929	0.5425	0.4405	0.2321	0.1689	0.0863	0.0293

The best values of γ, θ and δt are shown in italics and chosen on the basis of η.

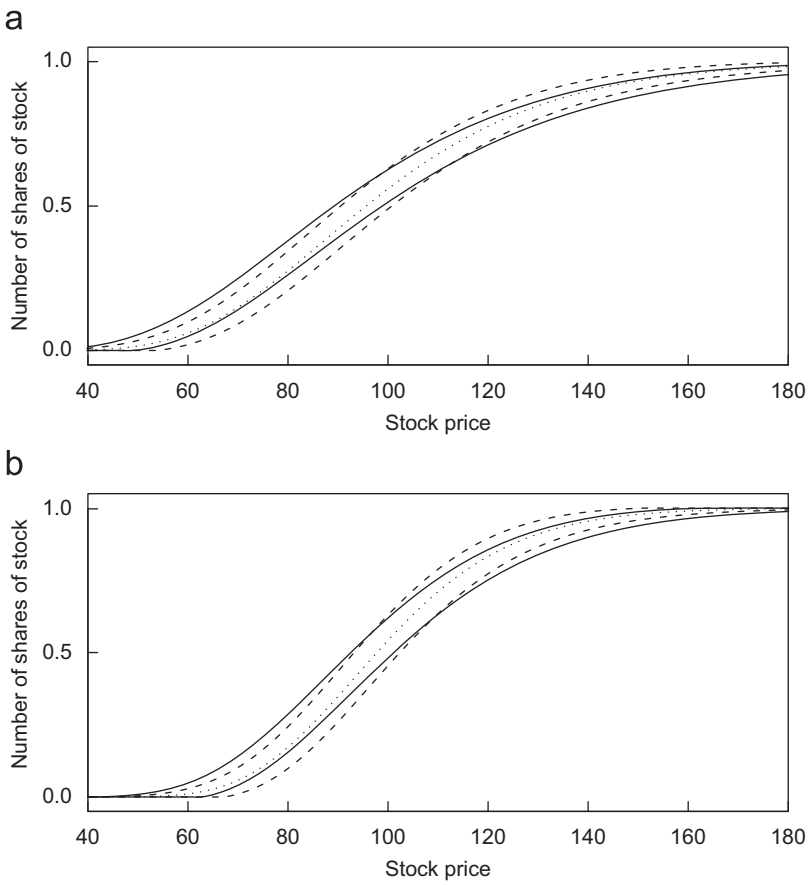

Fig. 3. Comparison of the buy–sell boundaries, as functions of the stock price S at (a) $T-t=1$ or (b) $T-t=0.25$, of the utility-maximized strategy with $\gamma=0.6$ (solid lines) with those of the cost-constrained risk-minimized strategy with $\theta=0.3$ (dashed lines). The boundaries are for a short call with $K=100$ and asset settlement with $\alpha=r=0$, $\sigma=0.3$ and $\lambda=\mu=0.01$. The Black–Scholes market deltas (dotted lines) are included for reference.

risk-minimized strategy, respectively. For the first scenario, Table 1 presents that η is minimized when we choose $\gamma=0.6$ for the utility-maximized strategy and $\theta=0.3$ for the risk-minimized strategy, yielding comparable minimum values of η (2.3190 and 2.3727, respectively). A comparison of the buy–sell boundaries of these two strategies in the first scenario is given in Fig. 3. Although the buy–sell boundaries of the utility-maximized strategy, unlike those of the risk-minimized strategy, are not (approximately) centered about the Black–Scholes delta (which is the market delta here under the first scenario), they bracket the Black–Scholes delta when the risk-aversion coefficient is chosen to be $\gamma=0.6$.

We have also included in Tables 1 and 2 values of the performance measures for the discrete-time Black–Scholes delta-hedging strategy and the discrete-time terminal-payoff replicating strategy of Leland (1985). Whereas Leland uses simultaneous adjustments of stock and bond positions at each revision date, resulting in a strategy that is not self-financing, only the stock position is rebalanced by using Leland's delta in the strategy labeled "Leland" in Tables 1 and 2. For a given revision interval δt, the Black–Scholes (resp. Leland) strategy for a short call ($i=s$) consists of holding $\Delta_{\mathrm{BS}}(t_n, S_{t_n}; \sigma)$ (resp. $\Delta_{\mathrm{BS}}(t_n, S_{t_n}; \hat{\sigma})$) shares of stock at each revision date $t_n = n\delta t$ ($n=0,1,\ldots,N=T/\delta t$), where $(\hat{\sigma}/\sigma)^2 = 1+\sqrt{2}(\lambda+\mu)/(\sigma\sqrt{\pi\delta t})$. For a long call ($i=b$), Δ_{BS} (with σ or $\hat{\sigma}$) is replaced by $-\Delta_{\mathrm{BS}}$. Table 1 presents that η is minimized in the first

scenario when we choose $\delta t = \frac{1}{24}$ (equivalent to rebalancing once every 15 days in a 360-day year) for the Black–Scholes strategy and $\delta t = \frac{1}{30}$ (once every 12 days) for the Leland strategy. For the second scenario, Table 2 presents that η is minimized when we choose $\delta t = \frac{1}{72}$ (once every 5 days) for the Black–Scholes strategy and $\delta t = \frac{1}{90}$ (once every 4 days) for the Leland strategy. The minimum values of η (respectively, 3.1198 and 2.9749 in the first scenario, and 1.7093 and 1.2330 in the second scenario) are substantially larger than those for the utility-maximized and cost-minimized strategies in both scenarios.

Let $p = P(0, S_0)$ (or $-P(0, S_0)$) for a short (or long) call, and use C to denote C^s (or C^b). Then by (45),

$$\mathrm{E}\{(\mathrm{err})^2\} = p^2 + e^{-2rT}\mathrm{E}C^2 - 2pe^{-rT}\mathrm{E}C.$$

Therefore, minimizing $\mathrm{E}\{(\mathrm{err})^2\}$ is equivalent to minimizing $\mathrm{E}C^2 - a\mathrm{E}C$ with $a > 0$. This is also equivalent to the Markowitz-type problem of minimizing $\mathrm{Var}(C)$ subject to a target value of $\mathrm{E}(C)$, by an argument similar to that in the second paragraph of Section 2.5.

4.4. Volatility adjustment and another comparison of CARA utility maximization and cost-constrained risk minimization

For CARA utility with constant risk aversion parameter γ, Zakamouline (2006, pp. 19–20) equates the *target delta*, which is the mid-point of the buy–sell boundaries, to the Black–Scholes delta with a volatility adjustment that is a function of the risk aversion parameter γ, stock price S and time to expiration τ, and has shown that the "increased volatility (after adjustment) makes the middle of the no-transaction region a flatter function of the stock price than the Black–Scholes delta." We can follow Zakamouline (2006) to introduce a volatility adjustment that depends on S, τ and the Lagrange multiplier θ in the cost-constrained risk-minimization approach. Writing the volatility adjustment as $\hat{\sigma} = c\sigma$, let (L_c, M_c) denote the optimal cost-constrained risk-minimization hedging strategy corresponding to the adjusted volatility $\hat{\sigma}$ and Lagrange multiplier θ; see (43). Minimize the standard deviation of the hedging cost (which is the same as the standard deviation of the hedging error) over c, i.e.,

$$\hat{c}(\theta, S, \tau) = \arg\min_c \mathrm{Var}\{C^s(L_c, M_c; 0, T) | S_0 = S, x_0 = 0\},$$

where Var is computed by Monte Carlo simulations from model (1) with the actual σ. For $\lambda = \mu = 0.02$, this procedure yields $\hat{c} = 1.3244$ for the example associated with Fig. 2 in Section 4.2.

Fig. 4 plots the mean versus the standard deviation of the hedging error (under the first market scenario described in Section 4.3) for (a) the Black–Scholes and Leland strategies (labeled 1 and 2, respectively), both with the revision interval δt

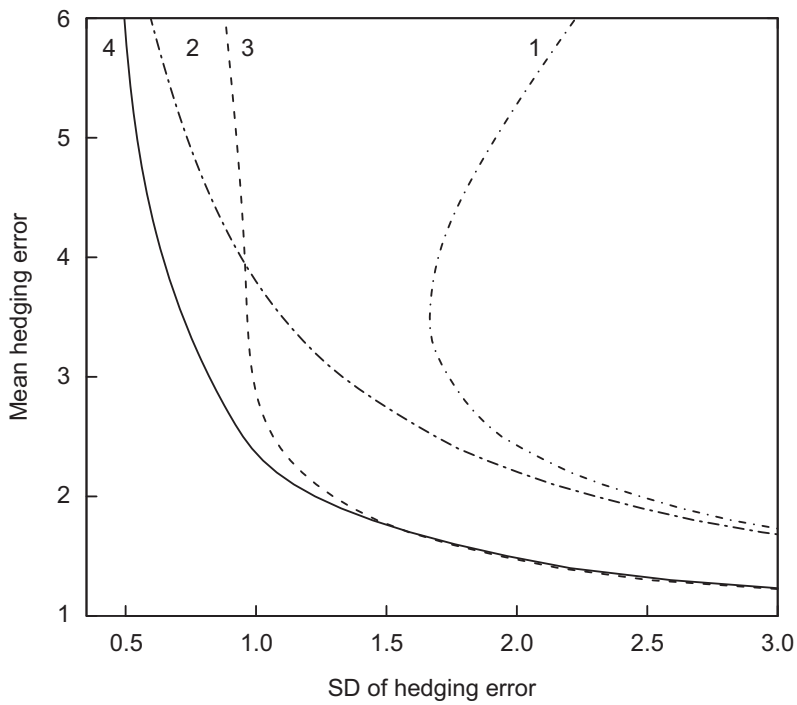

Fig. 4. Means and standard deviations (SD) of hedging error of four strategies for hedging a short call option under the Black–Scholes market price with $S = K = 100$, $T - t = 1$, $\alpha = r = 0$, $\sigma = 0.3$ and $\lambda = \mu = 0.01$. The strategies are labeled as (1) Black–Scholes (discrete-time pathwise replication using unadjusted delta), (2) Leland (discrete-time terminal-payoff replication using adjusted delta), (3) CARA utility maximization and (4) cost-constrained pathwise-risk minimization incorporating the volatility adjustment.

varying from 1 to 12 days, (b) the utility-maximization approach (labeled 3), with the risk-aversion coefficient γ of the negative exponential utility function $U(z) = 1 - e^{-\gamma z}$ varying from 0.1 to 100 and (c) the cost-constrained risk-minimization approach that incorporates volatility adjustment (labeled 4), with the Lagrange multiplier θ varying from 0.03 to 30. For the risk-minimized strategy with this volatility adjustment, not only does the standard deviation of the hedging error decrease while the mean hedging error increases with θ, but the mean versus standard deviation curve also lies to the lower left of the curves of the other three strategies when the standard deviation of the hedging error is less than 1.5 and is essentially the same as the utility-maximization curve for larger values of the standard deviation.

5. Concluding remarks

The preceding sections have focused on European call options on stocks that do not pay dividends. Since $\Delta^s(S) = -\mathbb{1}_{\{S<K\}}$ for a short put and $\Delta^b(S) = \mathbb{1}_{\{S<K\}}$ for a long put, Algorithms 1 and 2 can be applied to European put options by redefining $D^s(z) = -\mathbb{1}_{\{z<0\}}$ for a short put and $D^b(z) = \mathbb{1}_{\{z<0\}}$ for a long put. Recall also that for the put, $p_{\mathrm{BS}}(t,S;\sigma) = Ke^{-r(T-t)}\Phi(-d_2(t,S;\sigma)) - S\Phi(-d_1(t,S;\sigma))$ and $\Delta_{\mathrm{BS}}(t,S;\sigma) = -\Phi(-d_1(t,S;\sigma))$. When the stock pays dividends at the rate q, we can simply replace S in the preceding by $Se^{-q(T-t)}$. More generally, the algorithms can be modified in a straightforward manner to accommodate combinations of calls and puts (e.g., bull spreads, butterfly spreads, etc.) by changing the definitions of terminal wealth in (4) and (5).

In the presence of transaction costs, there is a tradeoff between minimizing the risk associated with writing an option (since the writer cannot hedge away the risk entirely by continuous trading of the underlying stock) and keeping the hedging cost at a minimum. The utility-maximization approach initiated by Hodges and Neuberger (1989) introduces a concave utility function (usually chosen to be of the CARA type for tractability and for its natural quantification of risk aversion) of terminal wealth and the formulation leads to the singular stochastic control problem in Section 2.2. We introduce a new approach to option hedging which is closer in spirit to the pathwise replication of Black and Scholes (1973) and the hedging-cost minimization of Bensaid et al. (1992). Our alternative cost-constrained pathwise-risk minimization formulation leads to the singular stochastic control problem in Section 2.3 which is simpler and for which an efficient backward induction algorithm (Algorithm 1) can be used to solve for the optimal buy and sell boundaries. We have also modified the algorithm to obtain a relatively simple algorithm (Algorithm 2) for utility-based option pricing and hedging.

Even though the utility maximization and the cost-constrained pathwise-risk minimization approaches use different notions of risk, we have shown in Section 4.3 that by choosing their associated parameters "optimally," both approaches have similar performance in terms of the root mean squared hedging error. Moreover, in Section 4.4 we have developed a volatility adjustment for the cost-constrained pathwise-risk minimization approach, which is analogous to that introduced by Zakamouline (2006) for the utility maximization approach. In this way and also in the computational algorithms of Section 3, a unified treatment of both approaches is provided.

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